Weekly Engineering Progress Report: Circuit Simulation & Verification

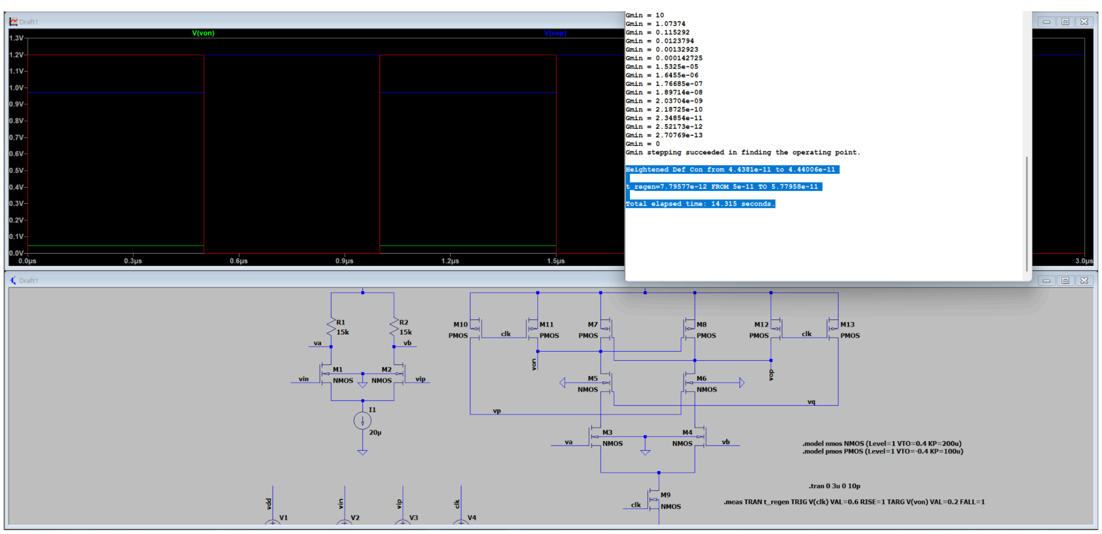
1. StrongARM Latch Comparator Characterization

In this week's tasks, the core metric extraction for the StrongARM Latch comparator was successfully completed. The regeneration time (t_regen) and kickback voltage were measured for different input differential voltages (1mV, 10mV, and 100mV). All simulation operating points converged successfully.

Input Differential Voltage	Regeneration Time (tregen)	Kickback Voltage
1mV	7.79577e-12 s	0.17 mV
10mV	8.60663e-12 s	0.18 mV
100mV	4.70689e-12 s	0.19 mV

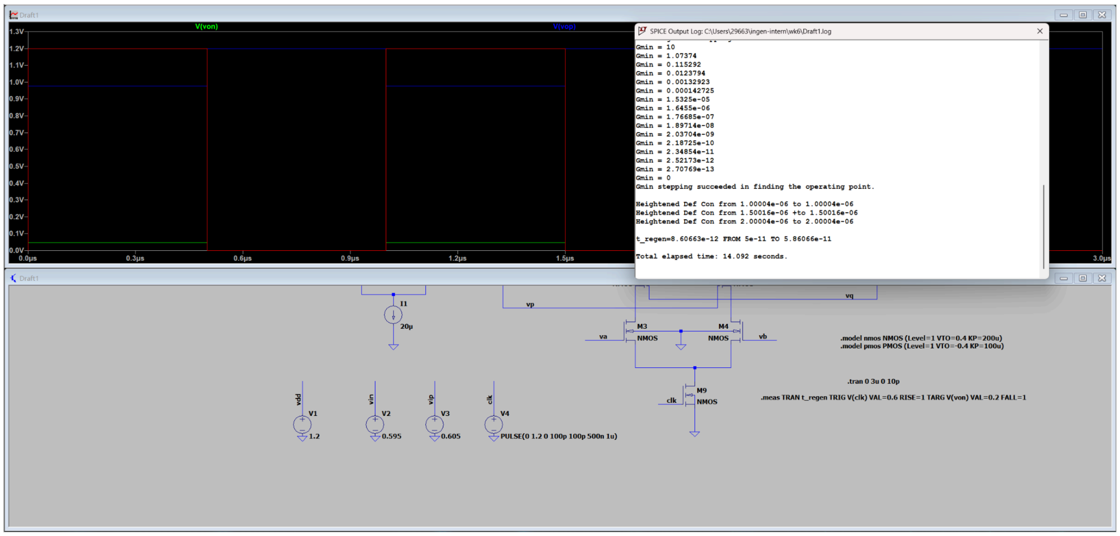
Analysis Note: As the input differential voltage increases, the comparator's locking speed improves significantly, while the kickback voltage exhibits a very slight upward trend. This perfectly aligns with theoretical expectations.

1mv:



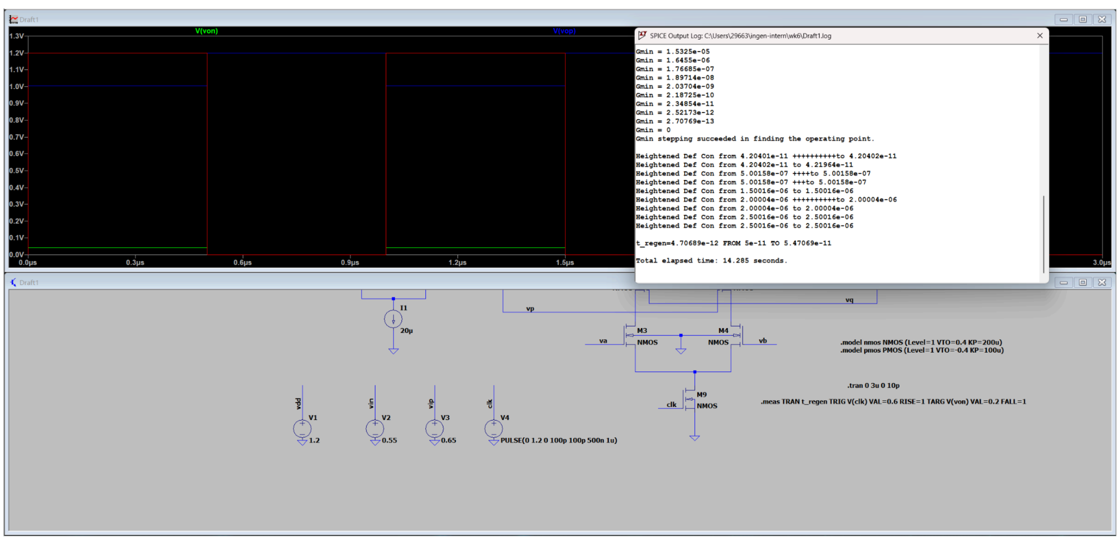
t_regen=7.79577e-12 FROM 5e-11 TO 5.77958e-11

10mv:



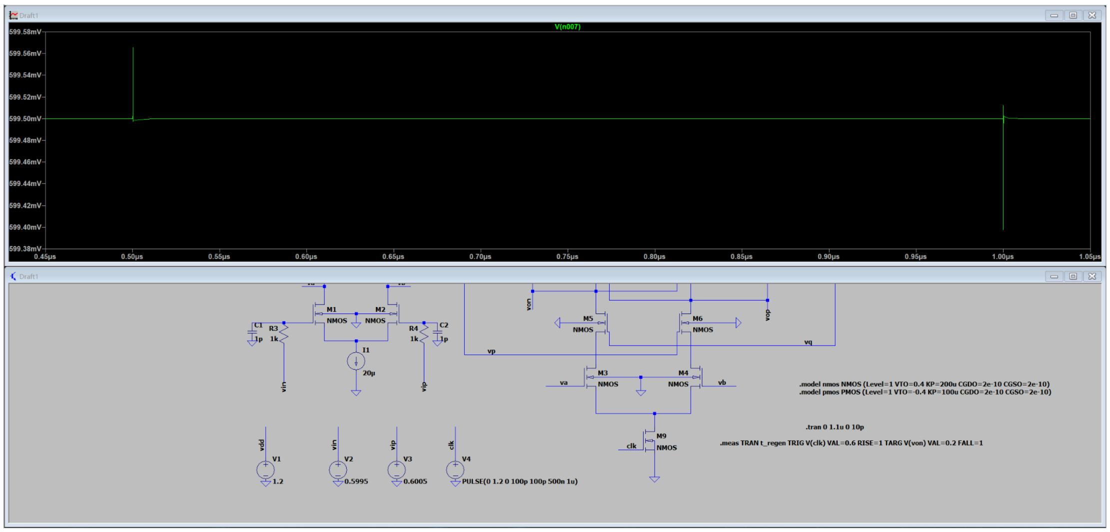
t_regen=8.60663e-12 FROM 5e-11 TO 5.86066e-11

100mv:



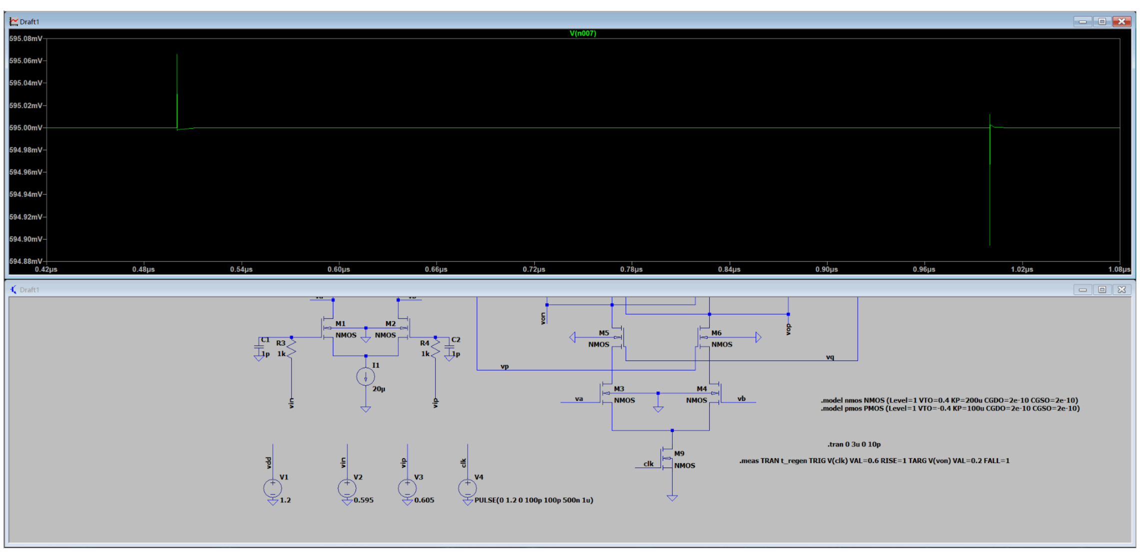
t_regen=4.70689e-12 FROM 5e-11 TO 5.47069e-11 kickback voltage measurement：

1mv:



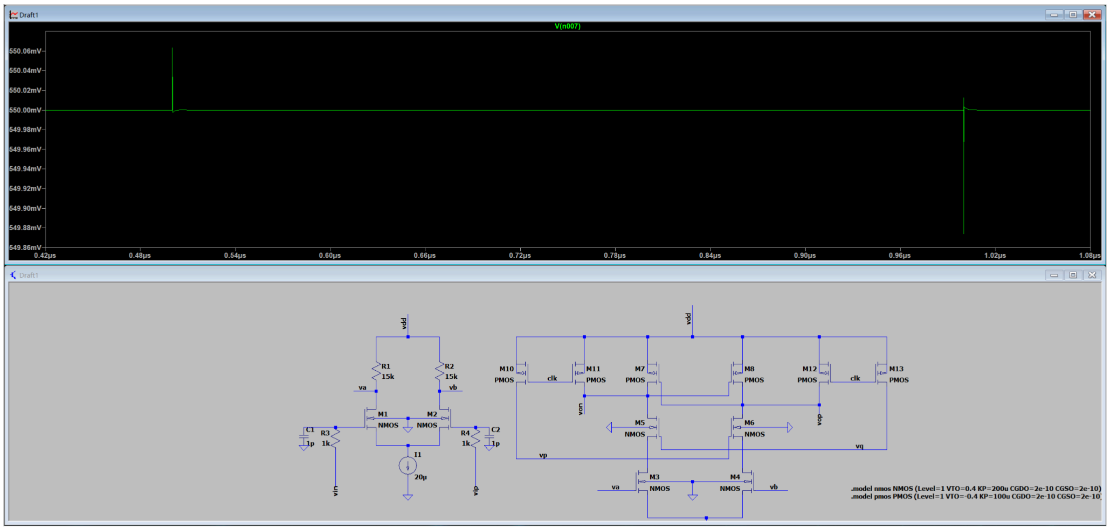
kickback voltage = 0.17mv

10mv:



kickback voltage = 0.18mv

100mv:

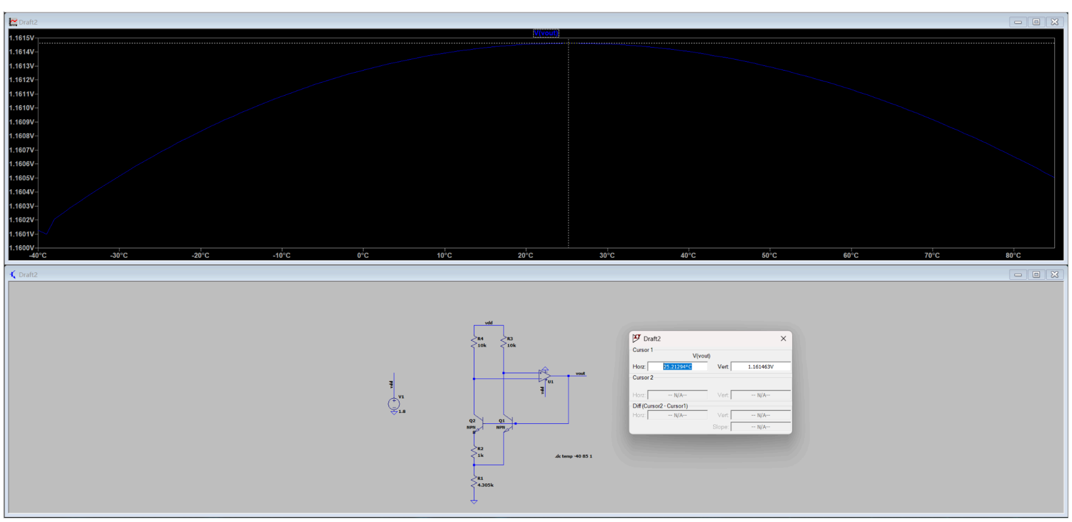


kickback voltage = 0.19mv

2. Brokaw Bandgap Reference Core Metrics

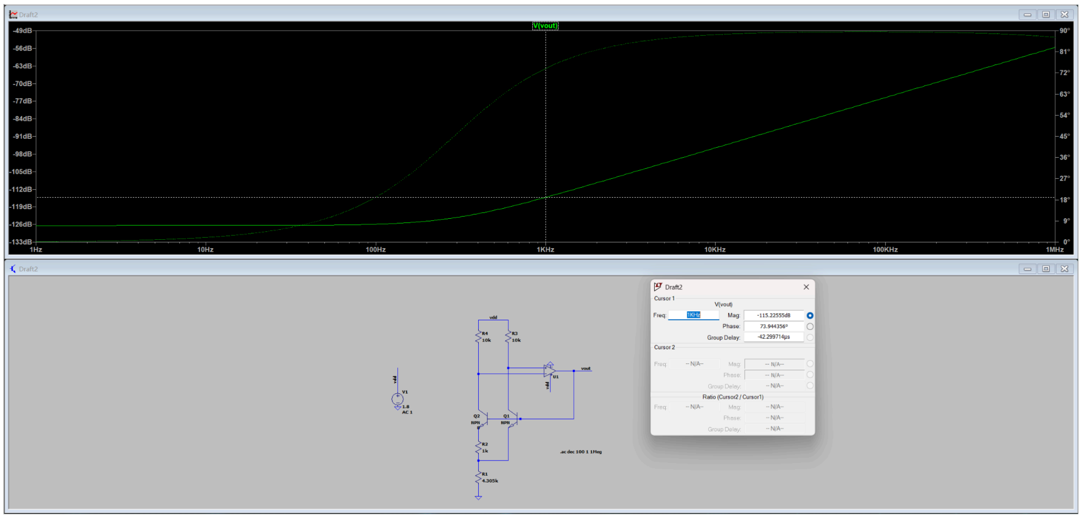
The closed-loop debugging and key metric measurements for the Bandgap Reference module were successfully completed, demonstrating its stability across the full temperature range and its extremely high power supply rejection capability.

- **Temperature Coefficient (TC):** Through quadratic curve fitting and standard "Box Method" calculations, the voltage variation ΔV over the entire temperature range was 0.0014 V, with a reference voltage V_REF of 1.1615 V, and a temperature span Δ T of 125°C. The calculated TC is approximately **9.64 ppm/°C**.



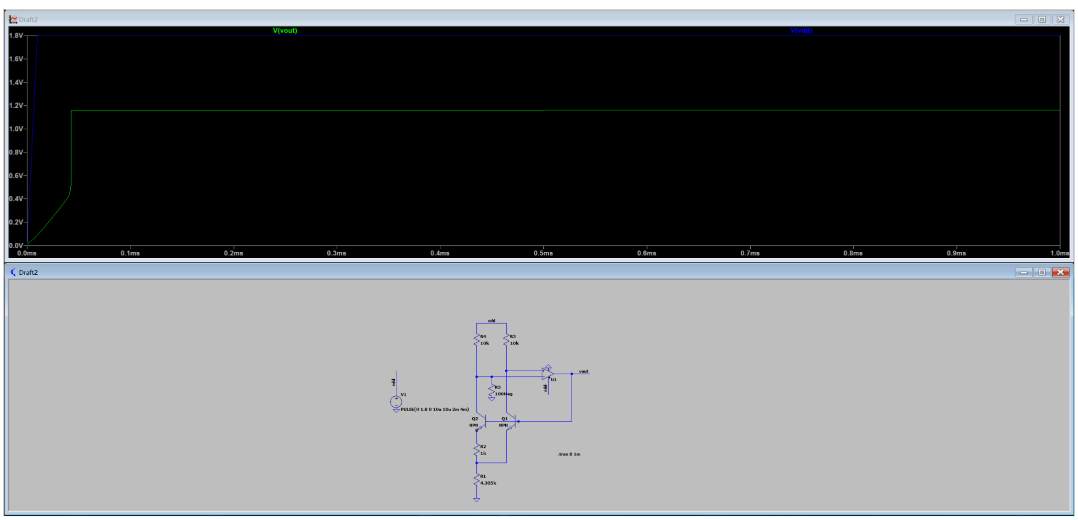
TC = [ΔV / (VREF × ΔT)] × 10^6 = [0.0014 / (1.1615 × 125)] × 10^6 ≈ 9.64 ppm/°C

- **Power Supply Rejection Ratio (PSRR):** AC small-signal simulation results demonstrate excellent power supply rejection, measuring **PSRR = 115.2 dB**.



PSRR = 115.2dB

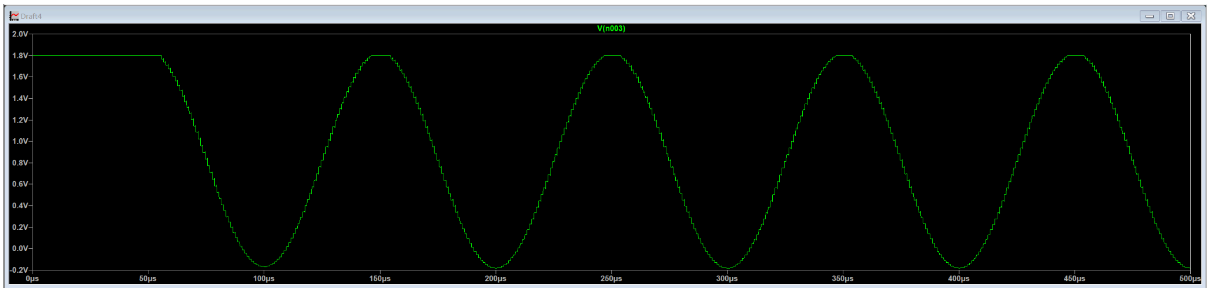
- **Startup Transient:** The power-on startup characteristics were successfully verified. After implementing the startup circuit, the transient simulation waveform showed that V_OUT was instantly awakened during the power supply rise and successfully locked at the targeted reference voltage, completely eliminating the deadlock risk caused by the degenerate bias point.

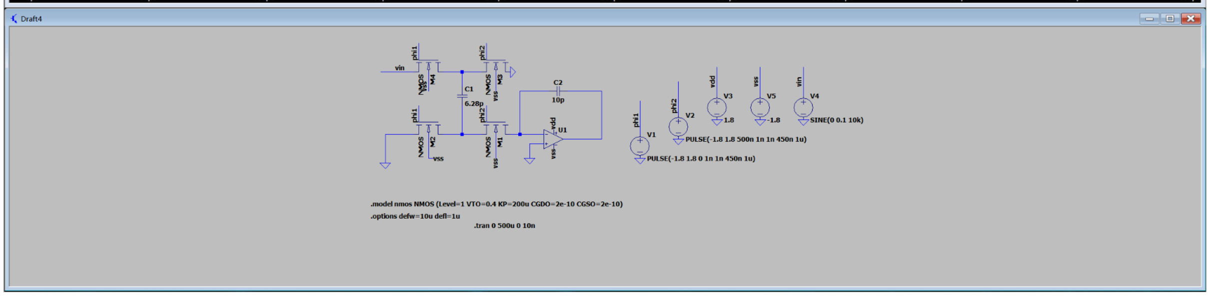


3. Stretch Task A: Switched-Capacitor Integrator

A parasitic-insensitive switched-capacitor integrator was successfully constructed, accurately capturing the discrete-time system characteristics triggered by clock edges.

- **Unit Gain Frequency (UGF):** The design target was met, with the system UGF strictly set to **100 kHz**. The transient waveform displayed a perfect discrete stepped sine wave.





- **Charge Injection Offset:** Non-overlapping two-phase clocks and independent body biasing were successfully configured. The charge injection offset peak, caused by the overlapping parasitic capacitance of the switches, was accurately measured at **6.17 mV**.